Lec-3


n = 5

```
Row=1   Star=1
while (row<n+n-2)
{ //star
  int i=1
  while (i<=star)
  {
    System.out.print("*");
    i++;
  }
  // new line
  System.out.println();
  row++;
  star++;
}
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;// user input
    int row = 1;
    int star = 1;
    int space = n - 1 - (n / 2);
    while (row < n + n - 1) {
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }
        // Mirror
        if (space >= 0) {// mirror
            space--;
        } else {
            star--;
        }
        // next row
        row++;
        System.out.println();
    }
}
```

N = 5

Row=1 Star=1 Space=2

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;// user input
    int row = 1;
    int star = 1;
    int space = n - 1 - (n / 2);
    while (row <= n + n - 1) {
        int i = 1;
        while (i <= star) {
            System.out.print("* ");
            i++;
        }
        // Mirror
        if (space >= 0) {// mirror wale line se pehle
            space--;
        } else {
            star--;
        }
        // next row
        row++;
        System.out.println();
    }
}
```

N = 5

Row=1 Star=1 Space=4

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

Ques 16:
n = 5

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

```
Row=1  1st Space=n-1,  Star=n
while (row<=n)
{
  //space
  //star
  if (row<=n)
  {
    space--;
    star++;
  }
  else if (row>n)
  {
    space++;
    star--;
  }
  row++;
}
```

n=5

```
*       *
* *     * *
* * *   * * *
* * * * * * * *
* * * *   * * *
* * *     * *
*       *
```

```
Row=1,  Star = n/2   Space = 1
while (row<=n)
{
  //space
  //star
  //mirror
  if (row<=n/2)
  {
    space++;
    star--;
  }
  else if (row>n/2)
  {
    space--;
    star++;
  }
  row++;
}
```

Ques 19:
n = 7

```
* * * * * * *
* * * * *
* * *
*
* * *
* * * * *
* * * * * * *
```

```
Row=1   Star= n/2   Space=n-1
while (row<=n)
{
  //star
  //space
  //mirror
  //n/2
}
```

```
public static void main(String[] args) {
    int n = 7;// user input
    int row = 1;
    int star = n / 2+1;
    int space = n - 1;
    int temp = 1;
    // int i = 1;
    while (row <= n) {
        int i = 1;
        while (i <= space) {
            System.out.print(" ");
            i++;
        }
        // space
        int j = 1;
        while (j <= star) {
            System.out.print("* ");
            j++;
        }
        // mirror
        if (row <= n / 2 + 1) {// mirror wale line se pehle
            space--;
            star += 2;
        } else {
            star+=2;
            space -= 2;
        }
        // next row prep
        row++;
        System.out.println();
    }
}
```

Row=1 Star=n Space=n-1

```
* * * * * * *
* * * * *
* * *
*
* * *
* * * * *
* * * * * * *
```

N = 5

Ques 25:
n = 5

```
        1
      2 3 4
    5 6 7 8 9
  10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    int n = 5;
    int row = 1;
    int star = 1;
    int space = n - 1 - 1;
    int val = 1;
    while (row <= n) {
        // space
        int i = 1;
        while (i <= space) {
            System.out.print(" ");
            i++;
        }
        // star
        int j = 1;
        while (j <= star) {
            System.out.print(val + " ");
            val++;
            j++;
        }
        // next row ki prep
        System.out.println();
        row++;
        star += 2;
        space--;
    }
}
```

N = 5

Row=1 Star=1 Space=4

```
        1
      2 3 2
    3 3 3 3 3
  2 2 2 2 2 2 2
1 1 1 1 1 1 1 1 1
```

Ques 27:
n = 5

```
1 <-
1 2 1
1 2 3 2 1 ->5
1 2 3 4 3 2 1 ->7
1 2 3 4 5 4 3 2 1 ->9
```

```
int val=1;
int i = 1;
while (i <= star) {
    System.out.print(val+" ");
    if(i<=star/2)
    {
      val++;
    }
    Else{
      val--;
    }
    i++;
}
```

```
public static void main(String[] args) {
    int n = 5;
    int row = 1;
    int star = 1;
    int space = n - 1;
    int val = 1;
    while (row <= n) {
        // space
        int i = 1;
        while (i <= space) {
            System.out.print(" ");
            i++;
        }
        // star
        int val = 1;
        int j = 1;
        while (j <= star) {
            System.out.print(val + " ");
            if (j <= star / 2 + 1) {
                val++;
            }
            else {
                val--;
            }
            j++;
        }
        // next row ki prep
        System.out.println();
        row++;
        star += 2;
        space--;
    }
}
```

Row=1 Star=1 Space=4

```
  1
1 2 1
1 2 3 2 1
```

N=3

```
  1
2 3 2
3 4 5 4 3
  2 3 2
    1
```

N=5

```
        1
      2 3 2
    3 4 5 4 3
  4 5 6 7 6 5 4
5 6 7 8 9 8 7 6 5
  4 5 6 7 6 5 4
    3 4 5 4 3
      2 3 2
        1
```

```
while (row <= n + n - 1)
{
  // space
  int i = 1;
  while (i <= space)
  {
    System.out.print("  ");
    i++;
  }
  // star
  int j = 1;
  while (j <= star)
  {
    System.out.print("* ");
    j++;
  }
  // mirror
  if (row <= n / 2)// starvator2-2;
  {
    space--;
  }
  else if (row > n / 2)// starvator2-2;
  {
    space++;
  }
  // next row ki prep
  row++;
  System.out.println();
}
```

N=5 Row=1 val=1 Space=4 Star=1

```
    1
  2 3 2
3 4 5 4 3
  2 3 2
    1
```

```
public static void main(String[] args) {
    // TODO Auto-generated method stub
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    int row = 1;
    int star = n - 1;
    int space = 1;
    int val = 1;
    while (row <= 2 * n - 1) {
        // space
        int i = 1;
        while (i <= space) {
            System.out.print("  ");
            i++;
        }
        // star
        int j = 1;
        while (j <= star) {
            System.out.print(val + " ");
            if (j <= star / 2) {
                val++;
            }
            else {
                val--;
            }
            j++;
        }
        // Mirror
        if (row <= n / 2) {
            space--;
            star += 2;
            val=1;
        }
        else {
            star -= 2;
            space++;
            val=1;
        }
        // Next line ki prep
        row++;
        System.out.println();
    }
}
```

N=3, Row=1 val=1 Space=1 Star=3

```
  1
1 2 1
3 4 5 4 3
```